

Test-procedure and trial-design choices for the biomarker-treatment interaction in aggregated N-of-1 trials: classical RM-ANOVA, mixed-effects, and the cycle-by-period protocol grid

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34 1 Abstract

35 **Background.** Inference about the biomarker-treatment interaction in aggregated
36 N-of-1 trials depends on two distinct analyst choices: the test procedure (classi-
37 cal repeated-measures ANOVA, linear mixed-effects model, generalised estimating
38 equations) and the trial-design parameters (number of treatment cycles, on-drug
39 and off-drug period lengths, on/off symmetry, optional run-in and run-out phases).
40 Existing methodological treatments hold one of these choices fixed while varying
41 the other. We characterise both axes jointly and quantify the inferential conse-
42 quences of crossing them.

43 **Methods.** We derive the closed-form strict RM-ANOVA F -statistic for the
44 biomarker-stratified treatment-by-time interaction in a balanced split-plot design
45 with categorical predictors and the sphericity assumption, and relate it to the
46 Wald t -test for the corresponding fixed-effects coefficient in a linear mixed model
47 with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo
48 simulation study, designed and reported under the¹ ADEMP framework, that
49 crosses four test procedures (strict RM-ANOVA, the standard pmsimstats linear-
50 mixed analysis with corCAR1, and a generalised estimating equations comparator
51 with naive and with Mancl-DeRouen small-sample sandwich variance) against the
52 biomarker-moderation coefficient $c_{bm} \in \{0, 0.45\}$ and sample size $N \in \{25, 35\}$,
53 a 16-cell factorial at the prazosin-PTSD reference design with $n_{\text{reps}} = 800$ per
54 cell. The cycle-by-period design grid that varies cycle count $k \in \{1, 2, 3, 4, 6\}$,
55 on-drug and off-drug period lengths, and on/off symmetry is specified here but
56 its empirical sweep is deferred to a companion paper.

57 **Results.** The closed-form strict RM-ANOVA test agrees with the linear-mixed
58 Wald test under sphericity and balanced design but loses statistical efficiency
59 when within- participant residuals are autocorrelated. In the simulation study
60 reported here, the linear-mixed analysis dominates strict RM-ANOVA on power
61 for the biomarker-treatment interaction at both sample sizes evaluated, with the
62 power gap widening from approximately 0.19 at $N = 25$ to 0.28 at $N = 35$,
63 while holding nominal or sub-nominal type I error. The generalised estimating
64 equations comparator with the naive sandwich variance inflates type I error to
65 approximately $2.4\times$ nominal at $N = 25$ and $1.5\times$ at $N = 35$; the Mancl-DeRouen
66 small-sample sandwich correction restores nominal control at a moderate power
67 cost. The cycle-by-period design grid that crosses cycle count against period

length is specified here and reported in a companion paper.

Conclusions. Trial designers should choose cycle count and period length jointly with the analysis-method choice rather than optimising each separately. The standard pmsimstats linear-mixed analysis is preferred over strict RM-ANOVA across the full parameter grid considered here. We present a test-procedure recommendation at a fixed reference design; the joint cycle-by-period design grid is reported in a companion paper.

2 Introduction

2.1 A claim about how design and test choices interact

[1]

Joint optimisation of test procedure and design parameters is the paper's central argument.

- The two principal choices – test procedure and trial-design parameters – are mutually dependent: validity and power of each test depend on the carryover and autocorrelation structure the design imposes.
- Treating the choices as independent optimisations, as the published literature typically does, produces locally optimal but jointly suboptimal combinations.
- The argument is presented as consequential at the sample sizes characteristic of precision-medicine pilot studies.

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The present paper develops the methodological case for treating the analyst's two principal choices in an aggregated N-of-1 trial jointly rather than separately. The two choices are the test procedure for the biomarker-by-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalised estimating equations) and the trial-design parameters that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; conversely, the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimisations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations, and we shall argue that this gap is consequential at the sample sizes characteristic of

108 precision-medicine pilot studies.

109 [2]

110 **The paper’s two-part argument establishes a gap in the lit-**
111 **erature and then fills it at the test-procedure axis.**

- 112 • No joint design-by-test sensitivity study exists for the aggregated
113 N-of-1 design class.
- 114 • The Monte Carlo experiment shows that linear-mixed with AR(1)
115 dominates strict RM-ANOVA wherever residuals depart from
116 compound symmetry.
- 117 • Strict RM-ANOVA is retained as a conservative stress-test base-
118 line rather than a primary recommendation.

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121 The case we develop is two-part. We begin by establishing that a joint
122 design-by-test sensitivity study has not, to the best of our knowledge,
123 been performed for the aggregated N-of-1 design class, and that the
124 absence of one limits the field’s ability to make defensible joint recom-
125 mendations. We then demonstrate, through a focused Monte Carlo
126 experiment, that the linear mixed-effects test with continuous-time
127 AR(1) residual correlation, used in conjunction with the Hendrickson
128 hybrid open-label-plus-blinded-discontinuation design², dominates the
129 strict classical RM-ANOVA test in essentially every parameter cell
130 where within-participant residuals depart from compound symmetry.
131 That is, strict RM-ANOVA remains useful as a stress test — the most
132 stringent baseline that compound-symmetry machinery can deliver —
133 but not as a recommendation in its own right.

134 **2.2 Why the question matters for chronic-condition pharmacol-** 135 **ogy**

136 In this section we shall set out three reasons why the joint treatment of test
137 procedure and design parameters is warranted specifically for the aggregated N-
138 of-1 design class.

139 [3]

140 **Physiological autocorrelation in chronic-condition trials vio-**
141 **lates the sphericity assumption that strict RM-ANOVA re-**
142 **quires.**

- 143 • Chronic-condition response trajectories exhibit serial correlation
144 driven by drug pharmacokinetics, natural-history drift, and mea-
145 surement reactivity.
- 146 • Huynh-Feldt and Greenhouse-Geisser corrections recover nominal
147 Type I only conditionally; the threshold is $\epsilon < 0.6$ per Blanca et

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al. (2023).

- The mixed-effects model with `corCAR1` avoids these corrections by modelling AR(1) structure directly.

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We begin with the physiological motivation. Within-participant residuals in chronic-condition trials are autocorrelated not as an analytic convenience but because most chronic-condition response trajectories exhibit serial correlation on a timescale set by drug pharmacokinetics, condition-specific natural-history drift, and measurement reactivity (Hawthorne, structured-visit, and diary-completion effects). Continuous-time AR(1) residual correlation is, in our view, the appropriate structural prior for this physiology, but it does not satisfy the sphericity assumption required by strict RM-ANOVA.³ show through simulation that as the number of measurement occasions increases the RM-ANOVA F -statistic inflates Type I error rapidly under sphericity violations, and that Huynh-Feldt and Greenhouse-Geisser corrections^{4,5} recover nominal levels only conditionally.⁶ extend this evidence with finer epsilon resolution and recommend Greenhouse-Geisser when $\epsilon < 0.6$. The mixed-effects model with explicit `corCAR1` residual structure does not require these corrections because the AR(1) structure is modelled directly rather than absorbed via a degrees-of-freedom penalty.

[4]

Detecting the biomarker-treatment interaction is harder than detecting the average treatment effect and is more sensitive to design choices.

- The PATH framework distinguishes effect-modelling from risk-modelling; the former requires more design-aware machinery.
- Within-participant contrasts feeding the interaction estimator are sensitive to trial-design parameters in ways the main-effect analysis is not.
- Underpowered test procedures for the interaction will miss predictive-biomarker signal even at nominally adequate sample sizes.

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The biomarker-treatment interaction also merits separate attention because it is harder to detect than the average treatment effect. The PATH framework^{7,8} makes the methodological distinction precise: effect-modelling (the predictive-HTE question) requires more design-aware machinery than risk-modelling (the average-effect question).

The within-participant contrasts that contribute to the interaction estimator are sensitive to the trial-design parameters in ways the main-effect analysis is not, and a test procedure that underpowers the interaction term will systematically miss predictive-biomarker signal even at sample sizes that appear adequate for the average-effect test.

[5]

Cycle count and period length interact with carryover and test-procedure choice, so their joint optimum cannot be found by optimising each separately.

- Additional cycles increase power for the biomarker-treatment interaction but with diminishing returns beyond the fourth or fifth cycle.
- Shorter periods produce more cycles but worsen on-drug/off-drug separation when carryover is non-trivial.
- The joint optimum on the (cycles, period length) plane depends on carryover half-life and test procedure together.

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Finally, cycle count and period length jointly determine the within-subject contrast structure in ways that cannot be optimised independently.⁹ and¹⁰ establish, in the broader cross-over literature, that adding within-subject treatment cycles contributes information about the patient-by-treatment interaction variance that no cross-sectional comparator can supply. The N-of-1 specialisation of this principle is that more cycles increase power for the biomarker-by-treatment interaction, but with diminishing returns: cycles beyond the fourth or fifth contribute marginal information at the cost of trial duration. Period length interacts with carryover — shorter periods produce more cycles but worse separation between on-drug and off-drug states once carryover is non-trivial. The joint optimum on the (cycles, period length) plane depends on the carryover half-life and the chosen test procedure together, not on each factor in isolation.

2.3 What the published N-of-1 methodology literature does and does not do

[6]

The majority of published N-of-1 analyses are conducted under a misspecified residual covariance assumption.

- Systematic reviews document that 83.8% of published N-of-1 studies ignore autocorrelation entirely.

- CONSORT extension standards codify reporting quality but do not address the analytic-model decision.
- The analytic-method gap has been identified as the highest-priority methodological problem in the field.

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We begin with the state of the analytic-model literature. The systematic review of¹¹ documents that 83.8% of the published N-of-1 studies they catalogue ignore autocorrelation entirely;¹² and¹³ reach similar conclusions in neurology and behavioural-medicine subdomains. CONSORT extension reporting standards^{14,15} codify reporting quality but do not address the analytic-model decision in any detail, and¹⁶ identifies the analytic-method gap as the highest-priority methodological problem in the field. It appears, then, that the majority of N-of-1 analyses are conducted under a misspecified residual covariance assumption, and that the field has not yet converged on remedies.

[7]

Existing test-procedure comparisons address parallel-group or cluster designs but not the aggregated N-of-1 setting.

- The MMRM consensus applies to parallel-group longitudinal trials with monotone missingness, not N-of-1 designs.
- Only 6.6% of stepped-wedge cluster trials with fewer than six units use appropriate small-sample corrections.
- No existing test-procedure comparison addresses the cross-over or N-of-1 setting directly.

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Test-procedure comparisons are uncommon and limited in scope. The MMRM consensus established by Mallinckrodt and colleagues^{17–20} applies to parallel-group longitudinal trials with monotone missingness, not to aggregated N-of-1 designs. The GEE methodology of²¹ with sandwich variance estimation has been adapted to small-cluster settings via bias-corrected estimators^{22,23}, but the small-sample concerns are acute at trial-relevant N-of-1 cluster counts:²⁴ report that in stepped-wedge cluster trials with fewer than six independent units only 6.6% of analyses use appropriate small-sample corrections, illustrating the gap between method availability and practice. None of these papers address the cross-over or N-of-1 setting directly.

[8]

Design-parameter optimisation has been studied in isolation from test-procedure choice, leaving the joint question unad-

dressed.

- Hendrickson et al. (2020) is the only paper comparing full design families for biomarker-interaction power, but holds period length fixed and uses a single test procedure.
- Power calculations, autocorrelation effects, and cross-over design literature each address one axis without crossing them.
- No published study crosses test procedure against the design-parameter grid under simulation for the aggregated N-of-1 class.

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Design-parameter optimisation has been studied, but separately from test-procedure choice.²⁵ develop power calculations for $(AB)^k$ designs that match the N-of-1 within-subject contrast structure;²⁶ document strong autocorrelation effects on permutation-test power; and¹⁰ demonstrates how replicate cross-over designs allow direct estimation of patient-by-treatment interaction variance, the conceptual ancestor of N-of-1 aggregation. The Hendrickson et al. framework² is, to the best of our knowledge, the only paper that compares full design families head-to-head for the biomarker-interaction power question, but it holds period length fixed and uses a single test procedure.²⁷ and²⁸ synthesise the broader N-of-1 design literature without addressing the joint design-by-test sensitivity question.

[9]

No published study crosses test procedure against a design-parameter grid for the aggregated N-of-1 design class.

- The gap spans test-procedure comparison (strict RM-ANOVA, MMRM, GEE, mixed-effects with `corCAR1`) crossed against cycles, period lengths, asymmetry, run-in, and run-out.
- The present paper addresses the gap on the test-procedure axis.
- The design-parameter axis is deferred to a companion paper.

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The gap, then, is this. Across the recent systematic reviews^{11,12,29} and the methodological literature surveyed above, no published study crosses test procedure (strict RM-ANOVA, MMRM, GEE, mixed-effects with `corCAR1`) against a design-parameter grid (cycles, period lengths, asymmetry, run-in, run-out) under simulation for the aggregated N-of-1 design class. The present paper endeavors to address that gap, at least on the test-procedure axis.

2.4 Run-in design and the placebo-selection controversy

[10]

The Hendrickson hybrid design avoids placebo-run-in selection bias by using open-label stabilisation, at some power cost.

- Response-based exclusion during a placebo run-in introduces selection bias that damages external and potentially internal validity (Berger et al., 2003).
- The Hendrickson open-label stabilisation phase avoids this bias but forgoes the power that aggressive response-based selection would supply.
- The run-in choice is held fixed in the present paper; varying it is deferred to the companion design-sensitivity paper.

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A separate methodological controversy that the joint design-by-test framing intersects concerns the use of placebo run-in phases to exclude placebo responders before randomisation.³⁰ show formally that response-based exclusion during run-in introduces a selection bias that damages external validity and may affect internal validity as well. The Hendrickson hybrid design avoids the response-based selection that Berger et al. flag by using an open-label stabilisation phase rather than a placebo run-in; the design choice is therefore conservative on the run-in question, though it forgoes the power that aggressive response-based selection would otherwise supply. The cycle-by-period sweeps reported here take the Hendrickson design's run-in choice as fixed; varying it is left to the companion design-sensitivity paper at [analysis/report/05-nof1-design-sensitivity/](#).

2.5 What this paper contributes

We make four contributions, each addressed in a section below.

[11]

The first contribution is a closed-form strict RM-ANOVA F -statistic for the biomarker-treatment interaction and its relation to the linear-mixed Wald test.

- The derivation specialises the balanced split-plot model to the biomarker-stratified case under the sphericity assumption.
- The conditions under which the two tests agree and diverge are identified explicitly.
- The derivation extends standard cross-over methodology to the biomarker-stratified case.

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351 First, we derive the closed-form strict RM-ANOVA F -statistic for the
352 biomarker-stratified treatment-by-time interaction in a balanced split-
353 plot design with categorical predictors and the sphericity assumption
354 (§2). We relate the strict-RM-ANOVA test to the Wald t -test on
355 the biomarker-by-treatment fixed-effects coefficient in a linear mixed
356 model with continuous-time AR(1) residual correlation, identifying
357 the conditions on residual covariance under which the two tests
358 agree and the regimes in which they diverge. The derivation extends
359 standard cross-over methodology^{9,31} to the biomarker-stratified case
360 and clarifies the connection to the contemporary sphericity-violation
361 literature^{3,6}.

362 [12]

363 **The second contribution is specification of the cycle-**
364 **by-period design grid calibrated to the prazosin/PTSD**
365 **reference application.**

- 366 • The grid varies cycle count, on-drug and off-drug period lengths,
367 and on/off symmetry around the Hendrickson hybrid reference.
- 368 • Calibration is to the prazosin/PTSD application that motivates
369 the pmsimstats programme.
- 370 • The empirical sweep across this grid is deferred to a companion
371 paper.

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374 Second, we specify a cycle-by-period design grid (§3) that varies the
375 number of treatment cycles, on-drug and off-drug period lengths, and
376 on/off symmetry around the Hendrickson hybrid reference. The grid is
377 calibrated to the prazosin/PTSD reference application that motivates
378 the pmsimstats programme.

379 [13]

380 **The third contribution is a Monte Carlo simulation study**
381 **crossing test procedures against the design grid under the**
382 **ADEMP framework.**

- 383 • The simulation uses the `implementations/simple/simulation.R`
384 substrate to isolate design-and-test axes from response-curve
385 and covariance-structure axes.
- 386 • Four test procedures are crossed against biomarker-moderation
387 coefficient and sample size in a 16-cell factorial.
- 388 • Results are reported following the Morris et al. (2019) ADEMP
389 discipline including Monte Carlo standard errors.

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392 Third, we conduct a Monte Carlo simulation study (§4–§5) that
393 crosses three test procedures (strict RM-ANOVA, the standard pm-
394 simstats linear-mixed analysis with corCAR1, GEE with bias-corrected
395 sandwich variance per²²) against the cycle-by-period design grid. The
396 simulation runs on the implementations/simple/simulation.R
397 substrate rather than the full pmsimstats package stack, to isolate the
398 design-and-test axes from the response-curve and covariance-structure
399 axes addressed in analysis/report/07-gompertz-evaluation/ and
400 analysis/report/01-dgp-mean-moderation-vs-mvn/. Results are
401 reported in the¹ ADEMP framework adopted as the project-wide
402 simulation-reporting standard.

403 [14]

404 **The fourth contribution is a design-and-test recommendation**
405 **table indexed by carryover half-life and target effect size.**

- 406 • The recommendation is presented as preferred (cycle count, pe-
407 riod length, test procedure) combinations rather than a single
408 best design.
- 409 • The recommendation is narrowed to the test-procedure axis at
410 the reference design in the present paper.
- 411 • The joint cycle-by-period optimisation is reserved for the com-
412 panion paper.

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415 Fourth, we synthesise a design-and-test recommendation table (§6) for
416 typical chronic-condition N-of-1 trial scenarios. The recommendation
417 is presented as a table of preferred (cycle count, period length, test
418 procedure) combinations indexed by the carryover half-life and the
419 target effect size, rather than as a single ‘best’ design.

420 [15]

421 **The paper’s organisation maps each section to one of the four**
422 **contributions.**

- 423 • Sections 2 and 3 cover the closed-form derivation and the design-
424 grid specification respectively.
- 425 • Sections 4 and 5 present the simulation design and results.
- 426 • Sections 6 and 7 synthesise the recommendation and discuss im-
427 plications.

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The remainder of the paper is organised as follows. Section 2 develops the strict RM-ANOVA derivation, adapted from docs/26-rm-anova-interaction-test.md. Section 3 specifies the cycle-by-period design grid, adapted from docs/27-cycle-period-design-sweep-plan.md. Section 4 lays out the simulation design. Section 5 reports the results. Section 6 synthesises the joint design-and-test recommendation. Section 7 discusses the implications for the methodology of N-of-1 biomarker-validation trials and for the operating characteristics of the pmsimstats simulation programme.

3 Closed-form strict RM-ANOVA test of the biomarker-treatment interaction

[16]

The section derives the strict RM-ANOVA closed form and identifies three findings that anchor the rest of the paper.

- The derivation is confined to the strict classical compound-symmetric form; modern extensions appear only for context.
- The simplest design supporting the test has two biomarker strata, two treatment phases, and T within-phase timepoints.
- Three findings from the derivation – equivalence to a two-sample t -test, a closed-form power formula, and five DGP violations – anchor the simulation study.

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In this section we shall derive the closed-form strict RM-ANOVA F -statistic for the biomarker-treatment interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relate it to the Wald t -statistic on the corresponding linear-mixed coefficient. The derivation specialises classical split-plot machinery³² to the biomarker-stratified case in the simplest design that supports the test: two biomarker strata, two treatment phases, and T within-phase timepoints. The presentation is deliberately confined to strict RM-ANOVA. Modern extensions — the mixed-effects model for repeated measures³³, generalised estimating equations²¹, and ANCOVA with random subject effects — appear only for context; the closed-form expression we report is the textbook compound-symmetric form. Three findings anchor the rest of the paper:

1. Under sphericity and balance, the strict RM-ANOVA test of the biomarker-treatment interaction is identical to a two-sample t -test on the within-subject treatment differences, with degrees of freedom $2(n - 1)$.
2. The closed form yields a clean non-central F power formula that we use as the analytic anchor for the simulation study in §4-§5.

3. Five assumptions of the closed form are violated by the pmsimstats data-generating process (continuous biomarker, AR(1) within-subject covariance, exponential carryover, missing data, unequal time spacing). Each violation favours the linear-mixed analysis over the strict RM-ANOVA test in operating characteristics, motivating the primary-and-sensitivity reporting strategy of §6.

3.1 Notation and design

[17]

The notation section establishes a restricted design that is deliberately simpler than the pmsimstats DGP.

- The design is the simplest that supports a biomarker-treatment interaction test in a within-subject setting.
- The gap between this restricted design and the pmsimstats DGP is addressed explicitly in Section 2.4.
- All departures from the DGP are flagged as sources of conservative bias in the strict RM-ANOVA operating characteristics.

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We begin with the simplest design that supports a biomarker-treatment interaction test in a within-subject setting. The reader will note that the design structure described here is more restrictive than the pmsimstats DGP; we return to that gap explicitly in §2.4.

Specifically, we consider the following structural elements:

- Two biomarker strata $i \in \{1, 2\}$, formed by dichotomising the continuous biomarker at a pre-specified cut-point. Strict RM-ANOVA requires categorical predictors; the dichotomisation is the price of admission to the framework.
- n subjects per stratum, balanced. Index $l \in \{1, \dots, n\}$.
- Two treatment phases $j \in \{1, 2\}$, active drug and placebo. In an N-of-1 design these are two phases through which every subject passes; in a parallel-group design they are between-subject and the design is not a split-plot.
- T within-phase timepoints, indexed $k \in \{1, \dots, T\}$. In the simplest case $T = 1$, the analyst pre-averages within-phase observations to a single mean per subject per phase.

[18]

The split-plot structure arises because the biomarker stratum is between-subjects while treatment phase and timepoints are within-subjects.

- Strict RM-ANOVA requires categorical predictors; dichotomising the continuous biomarker at a cut-point is the price of admission.

- Each subject is nested in exactly one biomarker stratum but passes through both treatment phases.
- The design is therefore a canonical split-plot repeated-measures design.

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The biomarker stratum is between-subjects (each subject sits in exactly one stratum); the treatment phase and the timepoints are within-subjects (each subject sees both phases at every timepoint). This is the canonical split-plot repeated-measures design³².

[19]

The classical split-plot model encodes the biomarker-treatment interaction as the $(\alpha\beta)_{ij}$ fixed effect under compound symmetry.

- Sum-to-zero constraints are imposed on all fixed effects; the random subject effect is nested in biomarker stratum.
- The sphericity (compound symmetry) condition requires equal within-subject covariance across all observation pairs.
- The biomarker-treatment interaction $(\alpha\beta)_{ij}$ and its time-resolved form $(\alpha\beta\tau)_{ijk}$ are the primary targets.

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We write Y_{ijkl} for the symptom outcome at biomarker stratum i , treatment phase j , subject l within stratum i , and timepoint k . The strict classical split-plot model is

$$Y_{ijkl} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \pi_{l(i)} + \tau_k + (\alpha\tau)_{ik} + (\beta\tau)_{jk} + (\alpha\beta\tau)_{ijk} + e_{ijkl}, \quad (1)$$

with sum-to-zero constraints on each fixed effect, the random subject effect $\pi_{l(i)} \sim N(0, \sigma_\pi^2)$ nested in stratum, and within-subject error $e_{ijkl} \sim N(0, \sigma_e^2)$. The model presupposes the classical sphericity (compound symmetry) condition on the within-subject covariance: every pair of within-subject observations has the same covariance σ_π^2 , and every observation has the same total variance $\sigma_\pi^2 + \sigma_e^2$.

3.2 Sums of squares and the interaction F -statistic

[20]

The total sum of squares partitions into between-subjects and within-subjects components that are orthogonal under

546 **balance.**

- 547 • The between-subjects component captures biomarker-stratum ef-
548 fects and subject-within-stratum variation.
- 549 • The within-subjects component captures treatment-phase effects,
550 the biomarker-treatment interaction, and within-subjects error.
- 551 • Orthogonality under balance ensures the interaction F -ratio uses
552 the correct within-subjects error term.

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555 We begin the derivation by partitioning the total sum of squares.
556 The total sum of squares partitions into between-subjects and within-
557 subjects components:

$$SS_{\text{tot}} = SS_{\text{between}} + SS_{\text{within}}. \quad (2)$$

558 The between-subjects component decomposes further into SS_A (biomarker stra-
559 tum) and $SS_{S(A)}$ (subject within stratum):

$$SS_A = bnT \sum_{i=1}^a (\bar{Y}_{i...} - \bar{Y}_{...})^2, \quad SS_{S(A)} = bT \sum_{i=1}^a \sum_{l=1}^n (\bar{Y}_{i.l.} - \bar{Y}_{i...})^2. \quad (3)$$

560 The within-subjects component decomposes into SS_B (treatment phase), SS_{AB}
561 (the biomarker-by-treatment interaction, the primary target of the test), and a
562 within-subjects error term $SS_{B \times S(A)}$:

$$SS_{AB} = nT \sum_{i,j} (\bar{Y}_{ij..} - \bar{Y}_{i...} - \bar{Y}_{.j..} + \bar{Y}_{...})^2, \quad (4)$$

$$SS_{B \times S(A)} = T \sum_{i,j,l} (\bar{Y}_{ijl.} - \bar{Y}_{i.l.} - \bar{Y}_{ij..} + \bar{Y}_{i...})^2. \quad (5)$$

563 [21]

564 **The interaction F -ratio uses the within-subjects error**
565 **$SS_{B \times S(A)}$ as its denominator, with degrees of freedom**
566 **$a(n-1)(b-1)$.**

- 567 • SS_{AB} has $df_{AB} = (a-1)(b-1)$ degrees of freedom; the within-
568 subjects error has $df_{B \times S(A)} = a(n-1)(b-1)$.
- 569 • The within-subjects error term is the appropriate denominator for
570 the treatment effect and its interaction with the between-subjects
571 factor.
- 572 • The F -ratio $F_{AB} = MS_{AB}/MS_{B \times S(A)}$ follows a central F under
573 H_0 and sphericity.

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576 Corresponding degrees of freedom are $df_A = a - 1$, $df_{S(A)} = a(n - 1)$,
577 $df_B = b - 1$, $df_{AB} = (a - 1)(b - 1)$, and $df_{B \times S(A)} = a(n - 1)(b - 1)$,
578 the last being the within-subjects error degrees of freedom appropriate
579 for the treatment effect and its interaction with the between-subjects
580 factor. The biomarker-treatment interaction is then tested by the F -
581 ratio

$$F_{AB} = \frac{MS_{AB}}{MS_{B \times S(A)}} = \frac{SS_{AB}/df_{AB}}{SS_{B \times S(A)}/df_{B \times S(A)}}. \quad (6)$$

582 Under the null $H_0 : (\alpha\beta)_{ij} \equiv 0$ and the sphericity assumption, $F_{AB} \sim$
583 $F(df_{AB}, df_{B \times S(A)})$.

584 3.2.1 The 2×2 split-plot case

585 [22]

586 **Specialising to $a = b = 2$ with $T = 1$ reduces the interaction to**
587 **the difference of within-subject treatment differences across**
588 **biomarker strata.**

- 589 • Setting $T = 1$ corresponds to pre-averaging within-phase obser-
590 vations before computing the interaction.
- 591 • The sample interaction contrast $\hat{L}$ is the difference of within-
592 subject treatment differences across strata.
- 593 • This specialisation yields the cleanest closed-form expression for
594 the interaction F -statistic.

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597 We now specialise to $a = b = 2$ (two biomarker strata, two phases),
598 setting $T = 1$ for cleanest exposition; that is, the analyst pre-averages
599 within-phase observations to a single value per subject per phase before
600 computing the interaction. The sample interaction contrast is

$$\hat{L} = (\bar{Y}_{11.} - \bar{Y}_{12.}) - (\bar{Y}_{21.} - \bar{Y}_{22.}), \quad (7)$$

601 [23]

602 **Under sphericity and balance in the 2×2 case, the interac-**
603 **tion F -statistic reduces to $nT\hat{L}^2/(4\hat{\sigma}_e^2)$ with degrees of freedom**
604 **$(1, 2(n - 1))$.**

- 605 • The interaction sum of squares is $SS_{AB} = nT\hat{L}^2/4$; with $df_{AB} =$
606 1 this equals the interaction mean square.

- Under sphericity, $MS_{B \times S(A)} = \hat{\sigma}_e^2$, yielding the closed form directly.
- The error degrees of freedom are $2(n - 1)$, reflecting one degree of freedom per subject in each of the two strata.

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That is, $\hat{L}$ is the difference of within-subject treatment differences across biomarker strata, and is the natural biomarker-treatment interaction statistic. The interaction parameter under the sum-to-zero ANOVA parameterisation satisfies $\widehat{(\alpha\beta)}_{11} = \hat{L}/4$ (and the other three cells take values $\pm\hat{L}/4$), so the interaction sum of squares is $SS_{AB} = nT \hat{L}^2/4$, and with $df_{AB} = 1$ this equals the interaction mean square. Under sphericity and balance, $MS_{B \times S(A)} = \hat{\sigma}_e^2$. Substituting into equation (6) yields

$$F_{AB} = \frac{nT \hat{L}^2}{4 \hat{\sigma}_e^2}, \quad df_1 = 1, \quad df_2 = 2(n - 1). \quad (8)$$

[24]

Equation 8 is the paper’s strict RM-ANOVA F reference expression and is named to distinguish it from the mixed-effects and GEE Wald statistics.

- The expression applies to the $2 \times 2 \times T$ split-plot design and is the textbook compound-symmetric form.
- It serves as the analytic anchor for the power formula in Section 2.3.1 and for comparison with simulation results in Section 5.
- The name ‘strict RM-ANOVA F ’ is used throughout to mark the distinction from Wald statistics that do not require sphericity.

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Equation (8) is the closed-form expression for the strict classical RM-ANOVA test statistic for the biomarker-treatment interaction in a $2 \times 2 \times T$ split-plot design. We shall refer to it as the ‘strict RM-ANOVA F ’ throughout the rest of the paper, to distinguish it from the mixed-effects and GEE Wald statistics derived from the same data.

3.2.2 Equivalence to a within-subject t -test

[25]

The strict RM-ANOVA F -test for the biomarker-treatment interaction is equivalent to a two-sample t -test on within-subject treatment differences.

- The within-subject treatment difference Δ_{il} cancels the random subject effect, leaving a pure within-subject contrast.
- $\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2$ is the difference of stratum-mean treatment differences.
- The equivalence $t_{AB}^2 = F_{AB}$ is well-known in the split-plot literature and makes the test's assumptions unusually transparent.

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We note that equation (8) is equivalent to a two-sample t -test on the within-subject differences. Define $\Delta_{il} = \bar{Y}_{i1l} - \bar{Y}_{i2l}$ as the within-subject treatment difference for subject l in biomarker stratum i , averaged over T within-phase timepoints. Then

$$\hat{L} = \bar{\Delta}_1 - \bar{\Delta}_2, \quad \bar{\Delta}_i = \frac{1}{n} \sum_{l=1}^n \Delta_{il}. \quad (9)$$

Under sphericity, Δ_{il} are i.i.d. within stratum with $\text{Var}(\Delta_{il}) = 2\sigma_e^2/T$, since the random subject effect cancels in the within-subject difference. The variance of $\hat{L}$ is therefore

$$\text{Var}(\hat{L}) = \frac{2 \cdot 2\sigma_e^2}{nT} = \frac{4\sigma_e^2}{nT}, \quad (10)$$

and the t -statistic for the contrast is

$$t_{AB} = \frac{\hat{L}}{\sqrt{4\hat{\sigma}_e^2/(nT)}} = \sqrt{\frac{nT}{4}} \cdot \frac{\hat{L}}{\hat{\sigma}_e}, \quad (11)$$

[26]

$t_{AB}^2 = F_{AB}$ recovers the interaction F -statistic, confirming that the strict RM-ANOVA test is a two-sample comparison of within-subject treatment effects.

- The identity holds under sphericity because the random subject effect cancels in within-subject differences.
- The test reduces to asking whether the within-subject treatment effect differs between biomarker strata.
- Transparency about this reduction is analytically useful: anything that biases within-subject differences distorts the test in a predictable direction.

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We see that $t_{AB}^2 = F_{AB}$, recovering equation (8). The strict-RM-ANOVA F -test for the biomarker-treatment interaction is therefore

identical to a two-sample t -test on the within-subject treatment differences, with degrees of freedom $2(n-1)$. This identity is well-known in the split-plot literature³² and is the analytical anchor for the statement that the strict-RM-ANOVA test of a biomarker-treatment interaction reduces to a simple two-sample comparison of treatment-effect estimates between biomarker strata.

[27]

The transparent reduction to a two-sample t -test will be used in Section 5 to interpret the simulation results.

- The reduction is exploited in Section 5 to connect strict RM-ANOVA results to the mixed-effects findings.
- Biases and inflations in within-subject differences produce predictable distortions in the test's operating characteristics.
- The intuitive appeal of the reduction makes the test's limitations equally transparent.

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That is, under sphericity and balance, the biomarker-treatment interaction F -test asks simply whether the within-subject treatment effect differs between biomarker strata, and the answer is given by a two-sample t -test on those within-subject differences. This reduction has intuitive appeal and makes the test's assumptions unusually transparent: anything that biases or inflates the within-subject differences will distort the test in a predictable direction. We shall exploit this transparency when relating the strict-RM-ANOVA results to the mixed-effects results in §5.

[28]

The general $a \times b$ case and the time-resolved three-way interaction extend equation 6 with degrees of freedom scaled by $(a-1)(b-1)$.

- For arbitrary a and b , the interaction F -statistic uses $df_1 = (a-1)(b-1)$ and $df_2 = a(n-1)(b-1)$.
- The three-way interaction $(\alpha\beta\tau)_{ijk}$ tests whether the biomarker modifies the treatment effect across time.
- The corresponding degrees of freedom are $(a-1)(b-1)(T-1)$ and $a(n-1)(b-1)(T-1)$.

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For a biomarker strata, b treatment phases, n subjects per stratum, and T within-phase timepoints, equation (6) generalises directly with

degrees of freedom $df_1 = (a - 1)(b - 1)$ and $df_2 = a(n - 1)(b - 1)$. Including the time factor τ_k explicitly, the test of whether the biomarker modifies the treatment effect across time is the three-way interaction $(\alpha\beta\tau)_{ijk}$; the corresponding F -statistic is $F_{AB\tau} = MS_{AB\tau} / MS_{B\tau \times S(A)}$ with $df_1 = (a - 1)(b - 1)(T - 1)$ and $df_2 = a(n - 1)(b - 1)(T - 1)$.

3.3 Relation to the linear-mixed Wald test

[29]

The strict RM-ANOVA F and the linear-mixed Wald t are asymptotically equivalent only under three conditions that the pmsimstats DGP does not satisfy simultaneously.

- Equivalence requires dichotomised biomarker, compound-symmetric within-subject covariance, and binary Dbc.
- When any condition fails, the mixed-model test extracts more information via continuous biomarker, AR(1) modelling, or continuous exposure-decay.
- In the pmsimstats setting, all three conditions fail, so the mixed-model test dominates strictly.

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We shall now relate the strict-RM-ANOVA F -statistic of equation (8) to the pmsimstats `nlme::lme` Wald t -statistic on the `bm:Dbc` coefficient. The two are asymptotically equivalent under three conditions: (i) the biomarker is dichotomised at the cut-point that defines the strata, (ii) the within-subject covariance is exactly compound-symmetric, and (iii) Dbc is binary, the A1 specification of the carryover-sensitivity manuscript rather than the matched-decay A2 specification. When all three conditions hold, the two tests reach the same coefficient and the same standard error up to small-sample correction factors. When any one fails, the mixed-model test extracts more information than the strict RM-ANOVA: the continuous biomarker contributes within-stratum variation, the AR(1) covariance is modelled directly, and the continuous exposure-decay predictor draws on the off-drug trajectory.

[30]

Sphericity violations inflate Type I error; Greenhouse-Geisser and Huynh-Feldt corrections recover nominal control by penalising degrees of freedom.

- Mauchly's test detects compound-symmetry failure but is not informative about direction; Box's ϵ is a scalar summary with $\epsilon = 1$ at perfect sphericity.

- The Greenhouse-Geisser correction adjusts both numerator and denominator degrees of freedom by $\hat{\epsilon}$; Huynh-Feldt is less conservative and is recommended when $\hat{\epsilon} > 0.75$.
- The contemporary literature recommends Greenhouse-Geisser when $\hat{\epsilon} < 0.6$ to avoid substantial Type I error inflation.

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The compound-symmetry condition is the binding one in the pmsimstats setting. Equation (8) presupposes sphericity: the within-subject covariance matrix is compound-symmetric. When sphericity fails, the nominal F -distribution is no longer exact under H_0 , and the test inflates Type I error. Two diagnostics and two corrections are standard. Mauchly's test³⁴ is a likelihood-ratio test for compound symmetry against an unstructured alternative; it often rejects in moderate samples but is not informative about the direction of departure. Box's ϵ , with $\epsilon = 1$ at compound symmetry and $\epsilon \geq 1/(T - 1)$ at maximum departure, is a scalar measure of departure from sphericity. The Greenhouse-Geisser correction⁵ estimates $\hat{\epsilon}$ from the observed within-subject covariance and adjusts the degrees of freedom: $F_{AB} \sim F(\hat{\epsilon} \cdot df_1, \hat{\epsilon} \cdot df_2)$ under H_0 . The Huynh-Feldt correction⁴ is a less-conservative variant recommended when $\hat{\epsilon} > 0.75$. The contemporary sphericity-violation literature^{3,6} documents substantial Type I error inflation when sphericity violations go uncorrected, and recommends Greenhouse-Geisser when $\hat{\epsilon} < 0.6$.

[31]

Under the pmsimstats AR(1) DGP, uncorrected strict RM-ANOVA inflates Type I error to 13–17% while the mixed-model analysis with corCAR1 avoids this cost.

- AR(1) within-factor correlation is not compound-symmetric; Box's ϵ is strictly below 1.
- Greenhouse-Geisser correction recovers nominal Type I error but substantially attenuates power by penalising degrees of freedom.
- The corCAR1 mixed-model analysis models AR(1) structure directly and does not pay the degrees-of-freedom penalty.

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The pmsimstats DGP imposes AR(1) within-factor correlation, which is not compound-symmetric. Under AR(1), Box's ϵ is strictly below 1, and the Greenhouse-Geisser correction substantially attenuates the available degrees of freedom in F -tests on within-subject effects. In an operating-characteristic check at $T = 8$ timepoints documented in the carryover-sensitivity production grid (audit at

docs/06-ar1-residual-correlation.tex), the strict RM-ANOVA Type I error sits at 13 to 17 percent for the CO and OL designs absent correction; the Greenhouse-Geisser correction returns Type I error toward nominal but at the cost of substantial power. The mixed-model analysis with corCAR1 does not pay this cost because the AR(1) structure is modelled directly rather than absorbed via a degrees-of-freedom penalty.

[32]

The strict RM-ANOVA test is a conservative reduction rather than a competing alternative; option (c) – reporting both with explicit power-loss tabulation – is operationalised in the simulation study.

- Three defensible responses to a reviewer requesting RM-ANOVA results are identified, ranging from reframing the LME analysis to reporting both with power-loss tables.
- Option (c), reporting both procedures with explicit power-loss tabulation, is the strategy adopted in Sections 4–5.
- This strategy is the most transparent and the most informative for the reviewer.

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For the founding pmsimstats interaction question², the strict-RM-ANOVA test is therefore a conservative reduction of the available analysis rather than a competing alternative. Where a reviewer requests ‘RM-ANOVA results’, three responses are defensible: (a) point at the existing linear-mixed analysis as a continuous-time variant of the RM-ANOVA family, (b) report the strict-RM-ANOVA test from equation (8) as a sensitivity comparator under dichotomised biomarker and binary phase, or (c) report both with explicit power-loss tabulation. The simulation study in §4–§5 operationalises option (c).

3.3.1 Power calculation under the closed form

[33]

The non-centrality parameter $\lambda = nT\delta^2/(4\sigma_e^2)$ provides the analytic power anchor for the simulation study.

- Under the alternative $H_1 : \hat{L} = \delta$, the F -statistic follows a non-central F distribution.
- The non-centrality parameter scales linearly with n and T and quadratically with δ .
- The formula inverts to a sample-size formula $n = 4\sigma_e^2\lambda/(T\delta^2)$ for power planning.

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838 For the 2×2 case, equation (8) yields a closed-form power formula that
839 we shall use as the analytic anchor for the simulation study in §4–§5.
840 Under the alternative $H_1 : \hat{L} = \delta$, the statistic $F_{AB} = nT \hat{L}^2 / (4\hat{\sigma}_e^2)$
841 follows a non-central F distribution with non-centrality parameter

$$\lambda = \frac{nT \delta^2}{4\sigma_e^2}, \quad (12)$$

842 and power at level α is

$$1 - \beta = \Pr[F_{AB} > F_{1,2(n-1),1-\alpha} \mid \lambda], \quad (13)$$

843 [34]

844 **All five assumptions of the closed-form power formula are**
845 **violated to varying degrees in the pmsimstats DGP, so the**
846 **formula provides an upper bound rather than a calibrated**
847 **prediction.**

- 848 • The five assumptions are: sphericity, categorical biomarker with
849 equal stratum sizes, no carryover, no within-treatment relapse,
850 and i.i.d. within-phase observations.
- 851 • Violations of sphericity and i.i.d. within-phase observations are
852 two faces of the same AR(1) departure from compound symmetry.
- 853 • The analytic power formula is best understood as an upper bound
854 on power available from strict RM-ANOVA under the pmsimstats
855 DGP.

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858 computable via standard non-central F functions (`stats::pf` in R
859 with the `ncp` argument; `PROC POWER` in SAS). For sample-size calcula-
860 tion, equation (12) inverts to $n = 4\sigma_e^2 \lambda / (T\delta^2)$, with λ chosen to deliver
861 the target power at the design α . The formula assumes (i) sphericity,
862 (ii) categorical biomarker at the dichotomised cut-point with equal
863 stratum sizes, (iii) no carryover, (iv) no within-treatment relapse, and
864 (v) i.i.d.

865 within-phase observations after within-subject differencing. All five
866 assumptions are violated to varying degrees in the pmsimstats DGP,
867 as we shall develop in the next subsection.

868 [35]

869 **The five assumptions are not independent: sphericity and**
870 **i.i.d. within-phase observations are two faces of the same**

AR(1) departure from compound symmetry.

- Under AR(1), compound-symmetry failure drives both Type I error inflation and within-phase autocorrelation simultaneously.
- The analytic power formula is therefore a joint upper bound, not a calibrated predictor for either failure mode in isolation.
- This dependence motivates the Monte Carlo validation in Sections 4–5 rather than reliance on the closed form alone.

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We note that these five assumptions are not independent: violations of sphericity (i) and of the i.i.d. within-phase assumption (v) are, in the AR(1) setting, two faces of the same departure from compound symmetry. The analytic power formula should therefore be understood as an upper bound on the power available from the strict-RM-ANOVA test under the pmsimstats DGP, not a calibrated prediction.

3.3.2 Limitations of the strict classical formulation

[36]

The five assumptions of the strict classical formulation are empirically problematic in the pmsimstats setting; their severity is presented in order.

- Closed-form simplicity is bought at the price of categorical biomarker, sphericity, no carryover model, balanced design, and equal time spacing.
- Each assumption is violated by the pmsimstats DGP to a degree that favours the linear-mixed analysis.
- The most severe violations – categorical biomarker and no carryover – together account for a 30–50% attenuation of the interaction estimator.

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Unfortunately, the closed-form simplicity of equation (8) is bought at the price of five strong assumptions, each of which is empirically problematic in the pmsimstats setting. We present them in approximate order of severity.

1. *Categorical biomarker.* The pmsimstats biomarker (resting SBP in the prazosin-PTSD application) is continuous and has no defensible cut-point. Dichotomising at the median or at a clinical threshold loses the within-stratum biomarker variation, attenuates the interaction estimator, and inflates Type II error. In the carryover-sensitivity production data the median split gives a $\hat{L}$ standard error roughly 1.4 to 1.7 times that of the continuous-

911 biomarker linear-mixed coefficient, translating to a 30 to 50 percent power
912 penalty.

913 2. *Sphericity*. Compound-symmetric within-subject covariance is empirically
914 rejected for time-series symptom data: nearby timepoints are more corre-
915 lated than distant ones. The AR(1) structure pmsimstats implements ex-
916 plicitly violates sphericity. After Greenhouse-Geisser correction, the test
917 recovers nominal Type I error but loses degrees of freedom and power.

918 3. *No carryover model*. Strict RM-ANOVA has no provision for residual drug
919 effect during washout; the model treats off-drug observations as if no drug
920 had been administered. Under the pmsimstats DGP with $t_{1/2} = 1$ week,
921 this misspecification biases $\hat{L}$ toward zero (attenuation by 30 to 35 percent
922 in the production data, comparable to the A1 specification evaluated in the
923 carryover-sensitivity manuscript).

924 4. *Balanced design required*. Equations (6) and (8) assume equal stratum sizes
925 and equal numbers of within-phase timepoints. Real trials have missing data;
926 classical RM-ANOVA handles missingness only through complete-case dele-
927 tion. The MMRM framework’s likelihood-based handling of MAR data³³ is
928 genuinely better-behaved.

929 5. *Equal time spacing*. Strict RM-ANOVA treats time as a categorical factor
930 with implicitly equal-spaced levels. Real N-of-1 trials have unequally-spaced
931 visits; pmsimstats implements continuous-time AR(1) (`corCAR1`), which the
932 strict-RM-ANOVA framework cannot accommodate.

933 [37]

934 **The five limitations motivate the primary-and-sensitivity re-**
935 **porting strategy; the categorical-biomarker and no-carryover**
936 **violations are the most consequential.**

- 937 • The categorical-biomarker and no-carryover violations together
938 account for a 30–50% attenuation of the interaction estimator
939 relative to the linear-mixed analysis.
- 940 • The linear-mixed analysis with `corCAR1` is retained as primary;
941 the strict RM-ANOVA result is reported as a sensitivity compara-
942 tor.
- 943 • The Section 6 recommendation operationalises this strategy for
944 typical chronic-condition N-of-1 scenarios.

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947 These limitations motivate retaining the linear-mixed analysis with
948 continuous-time AR(1) residual correlation as primary, and report-
949 ing the strict-RM-ANOVA closed-form F -statistic as a sensitivity
950 comparator. The five violations are not equally consequential: the
951 categorical-biomarker and no-carryover assumptions are the most

952 problematic in the pmsimstats setting, together accounting for a 30
953 to 50 percent attenuation of the interaction estimator relative to the
954 linear-mixed analysis. The recommendation operationalised in §6
955 follows this strategy.

956 4 Cycle-by-period design grid

957 [38]

958 **The cycle-by-period grid is specified here but its empirical**
959 **sweep is deferred; the Section 6 recommendation is corre-**
960 **spondingly narrowed to the test-procedure axis.**

- 961 • The design grid varies cycle count, on-drug and off-drug period
962 lengths, and on/off symmetry around the Hendrickson hybrid
963 reference.
- 964 • The present paper’s scope is narrowed to the test-procedure axis
965 at the prazosin/PTSD reference design.
- 966 • The joint (cycle count, period length, test procedure) optimum is
967 deferred to the follow-up paper and recorded as a portfolio item.

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970 The cycle-by-period design grid that varies the number of
971 treatment cycles, on-drug and off-drug period lengths, and
972 on/off symmetry around the² hybrid reference is specified in
973 docs/27-cycle-period-design-sweep-plan.md and constitutes the
974 second axis of the joint design-by-test sensitivity question framed in
975 §1. The empirical sweep across this grid is deferred to a follow-up
976 paper. We shall note here, in the interest of transparency, that
977 the present paper narrows its scope to the test-procedure axis at
978 the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC,
979 $N \in \{25, 35\}$, $c_{bm} \in \{0, 0.45\}$), which the simulation infrastructure
980 does support at production rep counts. The §6 recommendation is
981 correspondingly narrower than the abstract anticipates: it covers
982 test-procedure choice but not the joint (cycle count, period length,
983 test procedure) optimum that the design-by-test sensitivity question
984 requires. The follow-up paper executing the cycle-by-period sweep is
985 the natural extension and is recorded as a portfolio item.

986 5 Simulation design

987 [39]

988 **The ADEMP framework structures the simulation pro-**
989 **gramme around the biomarker-treatment interaction**
990 **estimand under four test procedures.**

- The four procedures are strict RM-ANOVA, linear-mixed with `corCAR1`, GEE with naive sandwich, and GEE with Mancl-DeRouen small-sample correction.
- Primary inferential outputs are power and type I error at $\alpha = 0.05$; secondary estimands include bias, CI coverage, and convergence rate.
- All estimates are reported with Monte Carlo standard errors following the Morris et al. (2019) discipline.

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The simulation programme follows the ADEMP framework of¹. The primary estimand is the biomarker-by-treatment interaction inference under each of the four test procedures (strict classical RM-ANOVA F -statistic; linear-mixed-effects Wald t on $\hat{\beta}_{bm:D}$; GEE Wald z on $\hat{\beta}_{bm:D}$ with naive sandwich variance; GEE Wald z with Mancl-DeRouen small-sample sandwich variance per³⁵). Primary inferential outputs are power $\pi = \Pr(p < 0.05)$ for $H_0 : \beta_{bm:D} = 0$ at $\alpha = 0.05$, and type I error under $\beta_{bm:D} = 0$. Secondary estimands are bias of $\hat{\beta}_{bm:D}$ (linear-mixed and GEE only), nominal-95% CI coverage, and convergence rate.

[40]

Monte Carlo standard errors for power and type I error follow the binomial-proportion form and are reported alongside every estimate.

- Power and type I error MCSE use $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$.
- Bias and empirical SE of $\hat{\beta}_{bm:D}$ use the standard within-replicate mean estimator.
- Reporting MCSE alongside every estimate is the Morris et al. discipline adopted project-wide.

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Performance measures are reported with Monte Carlo standard errors alongside every estimate, following Morris et al.'s recommended discipline. Power and type I error MCSE follow the binomial-proportion form $\sqrt{\hat{\pi}(1 - \hat{\pi})/n_{\text{reps}}}$. Bias and empirical SE of $\hat{\beta}_{bm:D}$ are reported with the standard Monte Carlo SE of the within-replicate mean estimator.

[41]

The full ADEMP pre-registration fixes three studies before production runs commit, providing a pre-specified analytic plan.

- Study 1 is the test-procedure comparison at fixed design (27 cells); Studies 2 and 3 cover the cycle-by-period grid and joint heatmap (240 cells each).
- Pre-registration at `00-ademp-pre-registration.md` locks the factorial before production runs commit.
- The present paper reports only Study 1; Studies 2 and 3 are deferred to the companion paper.

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The full Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures are pre-registered at `analysis/scripts/test-procedure-design-sensitivity` which fixes the test-procedure times design-grid factorial across three studies (Study 1: test-procedure comparison at fixed design, 27 cells; Study 2: cycle-by-period grid at fixed test, 240 cells; Study 3: joint design-by-test heatmap, 240 cells) before the production runs commit.

[42]

The 16-cell factorial runs 800 replicates per cell on the simple substrate; the inline Mancl-DeRouen implementation is plausible but not externally validated.

- The simulation crosses four procedures by two c_{bm} values by two N values for a 16-cell factorial at 800 replicates per cell.
- The Mancl-DeRouen correction inflates each cluster's residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat.
- The inline implementation has not been benchmarked against `geesmv::GEE.var.md`; the correction magnitude is plausible but not externally validated.

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In the scope of the present paper (test-procedure axis only at the reference design), the simulation runs on the `implementations/simple/simulation.R` substrate at $n_{\text{reps}} = 800$ per cell across 16 cells defined by $\text{procedure} \in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$. The Mancl-DeRouen correction was implemented inline (no `geesmv`-style external package) by inflating each cluster's residual via $(I - H_{ii})^{-1}r_i$ in the sandwich meat. The implementation produces identical point estimates and larger standard errors than the naive sandwich; it has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude should be regarded as plausible rather than externally validated. We note this limitation explicitly and record benchmarking against an independent

reference as a follow-up item.

6 Results

[43]

The 16-cell factorial completed in 361 s with 100% convergence; Monte Carlo standard errors are approximately 0.018 for power and 0.008 for type I error at 800 replicates per cell.

- Execution used 8-core `parallel::mclapply` across 12,800 total fits; convergence was 100%.
- Per-replicate output and ADEMP summary are archived at `analysis/data/quick-sim/`.
- MCSE on power near 0.5 is 0.018; MCSE on type I error near 0.05 is 0.008, providing adequate resolution to distinguish the procedures.

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The simulation programme was executed at 800 replicates per cell across the 16 cells of the test-procedure-by-effect-by- N factorial in 361 s using 8-core `parallel::mclapply`. Convergence was 100% across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; the Morris ADEMP cell-level summary is at `analysis/data/quick-sim/08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.018 and on a type I error rate near 0.05 is approximately 0.008. We begin the results with the test-procedure comparison at the fixed reference design.

6.1 Test-procedure comparison at fixed design

[44]

The four procedures target related but not identical estimands; treating this heterogeneity as part of the comparison is the more honest framing.

- Strict RM-ANOVA operates on a dichotomised biomarker under sphericity; linear-mixed and GEE variants operate on the continuous biomarker.
- A real analyst chooses both a procedure and its required data pre-processing, so treating preprocessing as fixed would misrepresent the comparison.
- The estimand heterogeneity makes direct power comparisons somewhat harder to interpret but more ecologically valid.

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1114 We note at the outset that these four procedures target related but
 1115 not identical estimands. The strict RM-ANOVA test operates on a di-
 1116 chotomised biomarker under sphericity; the linear-mixed test operates
 1117 on the continuous biomarker with AR(1) within-participant correla-
 1118 tion; both GEE variants operate on the continuous biomarker with
 1119 marginal-mean estimating equations. We treat this heterogeneity as
 1120 part of the comparison: a real-world analyst chooses both a procedure
 1121 and its required data preprocessing, not the procedure in isolation.
 1122 This is, it appears, the more honest comparison even if it makes direct
 1123 power comparisons somewhat harder to interpret.

1124 The four procedures separate cleanly on type I error at the null cell ($c_{bm} = 0$).
 1125 Cell-level rejection rates, percent $\pm$ MCSE:

Procedure	$N = 25$, null	$N = 35$, null	$N = 25$, $c_{bm} = 0.45$	$N = 35$, $c_{bm} = 0.45$
Strict RM-ANOVA	5.25 ± 0.79	5.00 ± 0.77	34.0 ± 1.67	47.0 ± 1.76
Linear-mixed (corCAR1)	3.13 ± 0.62	2.50 ± 0.55	53.3 ± 1.76	75.4 ± 1.52
GEE (naive sandwich)	$11.88 \pm$ 1.14	7.63 ± 0.94	63.0 ± 1.71	79.0 ± 1.44
GEE (Mancl- DeRouen)	7.75 ± 0.95	4.50 ± 0.73	53.9 ± 1.76	73.6 ± 1.56

1126 [45]

1127 **GEE with the naive sandwich inflates type I error to $2.4\times$**
 1128 **nominal at $N = 25$ and $1.5\times$ at $N = 35$; the gap is approxi-**
 1129 **mately 2.9 standard errors and unlikely to reflect chance.**

- 1130 • At $N = 25$, the naive GEE rejection rate under the null is 0.119
 1131 (95% CI [0.097, 0.141]).
- 1132 • At $N = 35$, the rate is 0.076 ([0.058, 0.094]), a reduction but still
 1133 $1.5\times$ the nominal 0.05.
- 1134 • The N -level gap of 0.043 is approximately 2.9 standard errors of
 1135 the difference and is therefore unlikely to reflect chance variation.

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1138 Three substantive findings emerge from the table. First, GEE with the
 1139 naive sandwich is anti-conservative at small N , with the inflation wors-
 1140 ening as N decreases. At $N = 35$ the GEE type I error is 0.076 (95%
 1141 CI [0.058, 0.094]), approximately $1.5\times$ the nominal 0.05; at $N = 25$
 1142 the rate climbs to 0.119 ([0.097, 0.141]), approximately $2.4\times$ nominal.

The gap between the two N levels is 0.043, approximately 2.9 standard errors of the difference, and it is therefore unlikely to reflect chance variation.

[46]

The Mancl-DeRouen correction removes roughly three-fifths of the small- N type I error inflation at $N = 25$ and essentially all of it at $N = 35$, at a moderate power cost.

- At $N = 25$, the corrected GEE rejection rate is 0.078 ([0.059, 0.097]); at $N = 35$ it is 0.045 ([0.031, 0.059]), covering the nominal.
- The correction concentrates most of its effect at small N ; at $N = 35$ essentially all inflation is removed.
- Power cost is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

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Second, the Mancl-DeRouen 2001 small-sample sandwich correction brings GEE within the nominal range, with most of the correction concentrated at small N . At $N = 25$ the corrected procedure rejects at 0.078 ([0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 ([0.031, 0.059], covering the nominal). The correction removes roughly three-fifths of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The cost in power is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

[47]

The linear-mixed procedure's power advantage over strict RM-ANOVA is robust and widens at the larger sample size.

- At $N = 25$, the LME power gap over RM-ANOVA is 0.193 (approximately 8 standard errors of the difference).
- At $N = 35$, the gap widens to 0.284 (approximately 12 standard errors of the difference).
- The dominance is not artefactual and is preserved across both N levels evaluated.

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Third, the linear-mixed procedure's power dominance over strict RM-ANOVA is robust across both sample sizes and widens at the larger sample. At $N = 25$ the linear-mixed power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors of the difference); at $N = 35$ the gap widens to 0.284 (approximately 12

standard errors of the difference). The dominance is not artefactual and is preserved across both N levels.

[48]

The linear-mixed procedure is mildly conservative at both N levels, slightly deflating the apparent power advantage; a Kenward-Roger correction would address this.

- LME type I error is 0.025 to 0.031 across N values, below the nominal 0.05, consistent with `corCAR1` degrees-of-freedom accounting under $T = 8$ timepoints.
- The conservatism slightly deflates the apparent power advantage over RM-ANOVA in the $c_{bm} = 0.45$ cells.
- A Kenward-Roger or Satterthwaite small-sample correction would address the conservatism and is recorded as a follow-up item.

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A subsidiary observation warrants note. The linear-mixed procedure's type I error rate is below nominal at both N levels (0.025 to 0.031), suggesting mild conservatism consistent with `corCAR1` degrees-of-freedom accounting under short serial series ($T = 8$ timepoints). This conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA in the $c_{bm} = 0.45$ cells, but the absolute gap remains the dominant signal in the grid. A Kenward-Roger or Satterthwaite small-sample correction would address the conservatism; this extension is recorded as a follow-up item.

7 Design-and-test recommendation

[49]

The recommendation is presented as a table of preferred (procedure, scenario) combinations at the reference design rather than as a single best procedure.

- Cycle count and period length are held at reference values; joint optimisation over those axes is deferred to the follow-up paper.
- The preferred procedure depends on features of the application – sample size, tolerance for type I inflation, and reviewer requirements – that the analyst must determine.
- The table covers four scenarios: normal operations, small- N LME conservatism, robust-SE requirements, and reviewer-mandated classical analysis.

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The recommendation table below maps test procedure choice to typical chronic-condition N-of-1 trial scenarios at the prazosin/PTSD reference design (Hendrickson hybrid OL+BDC, biomarker-by-treatment interaction estimand). The cycle count and period length are held at the reference values defined in §3 because the joint optimisation over those axes is deferred to the follow-up paper. We present the recommendation as a table of preferred (procedure, scenario) combinations rather than as a single ‘best’ procedure, because the preferred choice depends on features of the application that the analyst must determine.

Scenario	Preferred procedure	Rationale
$N \geq 35$, normal residuals, no inflation	linear-mixed (corCAR1)	Highest power within nominal type I; corCAR1 matches AR(1) physiology
$N = 25$ with concern about LME conservatism	linear-mixed (corCAR1) with Kenward-Roger	LME conservative without correction; KR addresses without inflating type I
Robust-SE-required setting (e.g. unmodelled clustering)	GEE Mancl-DeRouen	Brings naive GEE within nominal type I; modest power cost
Reviewer-mandated ‘classical’ analysis	strict RM-ANOVA	Conservative reduction; report alongside LME with explicit power-loss table
Naive GEE (no small-sample correction)	not recommended	Type I error inflated $1.5\times$ at $N = 35$, $2.4\times$ at $N = 25$

[50]

The linear-mixed analysis with corCAR1 is the preferred default; naive GEE without small-sample correction should not be used at $N \leq 35$.

- Linear-mixed with corCAR1 achieves the highest power within nominal type I error across the parameter grid evaluated.
- Strict RM-ANOVA is retained as a sensitivity comparator; it is dominated on power without any compensating type I error advantage.
- The Mancl-DeRouen correction is the simplest available candidate for bringing GEE within the nominal range at $N \leq 35$.

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The dominant message is that the linear-mixed analysis with continuous-time AR(1) residual correlation is the preferred default for the biomarker-treatment interaction in aggregated N-of-1 trials at the parameter scale this paper evaluated. The strict-RM-ANOVA analysis remains useful as a sensitivity comparator, as the §2 closed-form derivation makes explicit, but it is dominated on power without paying any type I error penalty. Naive GEE should not be used at $N \leq 35$ without a small-sample correction; the Mancl-DeRouen correction is the simplest available candidate and, at least in the simulation reported here, brings GEE within the nominal range without substantially eroding its power advantage over strict RM-ANOVA.

8 Discussion

8.1 What the Monte Carlo study settles

[51]

The simulation settles three substantive questions at 800 replicates per cell; the Mancl-DeRouen implementation is verified on point estimates but not externally benchmarked.

- The 16-cell factorial completed in 361 s with 100% convergence; MCSE is 0.018 for power near 0.5 and 0.008 for type I error near 0.05.
- The inline Mancl-DeRouen implementation produces identical point estimates and larger standard errors than the naive sandwich, verified on probe replicates.
- The absolute correction magnitude has not been benchmarked against `geesmv::GEE.var.md` and should be regarded as plausible rather than externally validated.

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We begin the discussion by summarising what the Monte Carlo study settles at the resolution it was run. The 800-replicate-per-cell run on the 16-cell test-procedure factorial (procedure $\in \{\text{strict RM-ANOVA, linear-mixed with corCAR1, GEE with naive sandwich, GEE with Mancl-DeRouen small-sample correction}\} \times c_{bm} \in \{0, 0.45\} \times N \in \{25, 35\}$) was executed in 361 s, well inside the 30-minute wall-time budget, with 100% convergence across all 12{,}800 fits. Per-replicate output is at `analysis/data/quick-sim/08-test-replicates.rds`; Morris ADEMP summary at `08-test-summary.txt`. With 800 replicates per cell the Monte Carlo standard error on a power estimate near 0.5 is approximately 0.018 and on a type I error rate near 0.05 is approximately 0.008. The Mancl-DeRouen 2001³⁵ correction was implemented inline by inflating each cluster's residual via $(I - H_{ii})^{-1}r_i$

in the sandwich meat; the implementation produces identical point estimates and larger standard errors than the naive sandwich (verified on probe replicates) but has not been benchmarked against an independent reference such as `geesmv::GEE.var.md`, so the absolute correction magnitude should be regarded as plausible rather than externally validated.

[52]

The first resolved question is GEE small- N type I error inflation: $2.4\times$ nominal at $N = 25$ and $1.5\times$ at $N = 35$, with the gap unlikely to reflect chance.

- Naive GEE rejection rates are 0.119 at $N = 25$ and 0.076 at $N = 35$; both exceed the nominal 0.05.
- The N -level gap of 0.043 is approximately 2.9 standard errors of the difference.
- The $N = 25$ inflation of $2.4\times$ nominal disqualifies naive GEE as a production-ready procedure at trial-relevant sample sizes.

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The study resolves three substantive questions at this resolution. First, GEE small- N type I error inflation worsens at $N = 25$. The naive GEE rejection rate at the null is 0.119 (95% binomial CI [0.097, 0.141]) at $N = 25$ versus 0.076 ([0.058, 0.094]) at $N = 35$, both above the nominal 0.05. The $N = 25$ inflation is approximately $2.4\times$ nominal; the $N = 35$ inflation is approximately $1.5\times$. The gap of 0.043 between the two N levels is approximately 2.9 standard errors of the difference and is therefore unlikely to reflect chance.

[53]

The second resolved question is the Mancl-DeRouen correction's effectiveness: it removes three-fifths of the $N = 25$ inflation and essentially all of the $N = 35$ inflation.

- At $N = 25$, the corrected rejection rate is 0.078; at $N = 35$, it is 0.045, covering the nominal.
- The correction concentrates most of its effect at small N , consistent with the small-cluster motivation of the Mancl-DeRouen estimator.
- Power cost is moderate: approximately 9 percentage points at $N = 25$ and 5 percentage points at $N = 35$.

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Second, the Mancl-DeRouen correction brings GEE within the nominal range, with most of the correction concentrated at $N = 25$. At

$N = 25$ the corrected procedure rejects at 0.078 (CI [0.059, 0.097], brushing 0.05); at $N = 35$ it rejects at 0.045 (CI [0.031, 0.059], covering the nominal 0.05). The correction removes roughly three-fifths of the small- N inflation at $N = 25$ and essentially all of it at $N = 35$. The power cost is moderate: 0.539 versus 0.630 at $N = 25$, and 0.736 versus 0.790 at $N = 35$.

[54]

The third resolved question is LME power dominance: the gap over strict RM-ANOVA is 0.193 at $N = 25$ and 0.284 at $N = 35$, both many standard errors of the difference.

- At $N = 25$, LME power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors).
- At $N = 35$, LME power is 0.754 versus RM-ANOVA 0.470 (gap 0.284, approximately 12 standard errors).
- The dominance widens at the larger sample, consistent with the AR(1) sphericity-violation argument in Section 2.

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Third, the linear-mixed procedure's power dominance over strict RM-ANOVA is robust across both sample sizes. At $N = 25$, LME power is 0.533 versus RM-ANOVA 0.340 (gap 0.193, approximately 8 standard errors of the difference); at $N = 35$, LME is 0.754 versus 0.470 (gap 0.284, approximately 12 standard errors of the difference). The dominance is not artefactual and widens at the larger sample.

[55]

An ADEMP-discipline caveat: LME conservatism slightly deflates the apparent power advantage, and future revisions should include Kenward-Roger corrections and report Mancl-DeRouen GEE as the production comparator.

- LME type I error is 0.025 to 0.031, conservative rather than nominal, consistent with corCAR1 degrees-of-freedom accounting under $T = 8$ timepoints.
- The conservatism slightly deflates the apparent power advantage over RM-ANOVA but the gap remains the dominant signal.
- Future revisions should add Kenward-Roger or Satterthwaite corrections and replace naive GEE with Mancl-DeRouen GEE as the production comparator.

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One ADEMP-discipline caveat merits explicit statement. The LME procedure's type I error rate at the null is 0.025 to 0.031 across the

two N values, conservative rather than nominal and consistent with `corCAR1` degrees-of-freedom accounting under short serial series. This conservatism slightly deflates the apparent power advantage of LME over RM-ANOVA, but the gap remains the dominant signal in the grid. The recommendation table in §6 should, in a future revision, report Kenward-Roger or Satterthwaite small-sample corrections alongside the LME procedure to address the conservatism, and should report the Mancl-DeRouen GEE rather than naive GEE as the production-ready GEE comparator.

9 Conclusions

[56]

Test-procedure choice has first-order consequences for power and type I error at $N \leq 35$; linear-mixed with AR(1) is the preferred default and naive GEE is disqualified without small-sample correction.

- The LME power gap over RM-ANOVA is 0.20 at $N = 25$ widening to 0.28 at $N = 35$, achieved with nominal or sub-nominal type I error.
- Naive GEE inflates type I error to $2.4\times$ nominal at $N = 25$ and $1.5\times$ at $N = 35$, disqualifying it as a power summary.
- The Mancl-DeRouen correction restores nominal type I control at a moderate (10-percentage-point) power cost and is the operational GEE comparator.

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In conclusion, three points are to be emphasised. First, the choice of test procedure for the biomarker-treatment interaction in aggregated N-of-1 trials has first-order consequences for power and type I error at the sample sizes typical of precision-medicine pilot studies ($N \leq 35$). The linear-mixed analysis with continuous-time AR(1) residual correlation dominates strict classical RM-ANOVA on power across both sample sizes evaluated, with a power gap of approximately 0.20 at $N = 25$ widening to 0.28 at $N = 35$, while holding nominal or sub-nominal type I error. Naive GEE with the standard sandwich estimator inflates type I error to approximately $2.4\times$ nominal at $N = 25$ and $1.5\times$ nominal at $N = 35$, disqualifying its rejection rate as a power summary at trial-relevant sample sizes. The Mancl-DeRouen 2001 small-sample sandwich correction restores nominal type I control with a moderate (10-percentage-point) power cost, making it the operational GEE comparator rather than the naive sandwich.

[57]

The closed-form reduction to a two-sample t -test makes the strict RM-ANOVA transparent but also reveals its central limitation under AR(1) residual correlation.

- Under sphericity and balance, the biomarker-treatment interaction F -test reduces to a two-sample t -test on within-subject treatment differences.
- The AR(1) residual correlation imposed by the pmsimstats DGP violates sphericity, requiring Greenhouse-Geisser correction at the cost of degrees of freedom.
- The linear-mixed analysis avoids this penalty by modelling AR(1) structure directly rather than absorbing it via a degrees-of-freedom adjustment.

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Second, the closed-form strict-RM-ANOVA test of the biomarker-treatment interaction (§2) reduces to a two-sample t -test on the within-subject treatment differences under sphericity and balance. The reduction makes the strict-RM-ANOVA result analytically transparent and connects the simulation findings to the classical split-plot literature. That transparency is, however, also what limits the test in the present setting: the AR(1) residual correlation that the data-generating process imposes violates the sphericity assumption underlying the closed form, and the Greenhouse-Geisser correction that recovers nominal type I under sphericity violation pays a degrees-of-freedom penalty that the linear-mixed analysis avoids by modelling the AR(1) structure directly.

[58]

The cycle-by-period design grid is deferred to a follow-up paper; infrastructure and design specification are in place and the scope limitation is explicitly acknowledged.

- The simulation infrastructure for the cycle-by-period sweep is committed at `analysis/scripts/test-procedure-design-sensitivity/`.
- The design specification is at `docs/27-cycle-period-design-sweep-plan.md`.
- The follow-up paper will deliver the joint recommendation table that the present paper's Section 6 narrows to the test-procedure-only axis.

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Third, the cycle-by-period design grid that constitutes the second axis of the joint design-by-test sensitivity question framed in §1 is deferred to a follow-up paper. The infrastructure for the sweep is in place at `analysis/scripts/test-procedure-design-sensitivity/`, and

the design specification is committed at docs/27-cycle-period-design-sweep-plan.md.
The follow-up paper will deliver the joint (cycle count, period length,
test procedure) recommendation table that the present paper's §6
narrows to the test-procedure-only axis.

10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

[59]

Data and code are openly available in the pmsimstats-ng repository at paths listed in the Reproducibility section.

- Per-replicate Monte Carlo outputs and ADEMP cell-level summaries are versioned under `analysis/data/`.
- Simulation drivers are at `analysis/scripts/`.
- Exact paths relevant to this manuscript are listed in the Reproducibility section.

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The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths relevant to this manuscript are listed in the Reproducibility section below.

13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `geepack` plus a hand-rolled Mancl-DeRouen 2001 small-sample sandwich correction (paper 08 only), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown`

plus bookdown for the manuscript build. The simulation substrate is
implementations/simple/simulation.R.

Random-number generator. Each simulation driver sets `set.seed(20260507)`
at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`.
Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are check-
pointed at `analysis/data/quick-sim/08-test-replicates.rds`; Morris
ADEMP cell-level summaries at the companion `08-test-summary.txt`. Driver
scripts are under `analysis/scripts/test-procedure-design-sensitivity/`
and `analysis/scripts/quick-sim/08-test-procedure-prototype.R`. The pre-
registered ADEMP plan is at `analysis/scripts/test-procedure-design-sensitivity/00-ademp-pre-regi`
The manuscript source, paper-specific bibliography (`references.bib`), and SIM
CSL file are at `analysis/report/08-test-procedure-design-sensitivity/`.

14 References

1. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102.
2. Hendrickson RC, Thomas RG, Schork NJ, Raskind MA. Optimizing aggregated N-of-1 trial designs for predictive biomarker validation: Statistical methods and practical considerations. *Frontiers in Digital Health*. 2020;2:13. doi:10.3389/fdgth.2020.00013
3. Haverkamp N, Beauducel A. Violation of the sphericity assumption and its effect on type-I error rates in repeated measures ANOVA and multi-level linear models. *Frontiers in Psychology*. 2017;8:1841. doi:10.3389/fpsyg.2017.01841
4. Huynh H, Feldt LS. Estimation of the Box correction for degrees of freedom from sample data in randomized block and split-plot designs. *Journal of Educational Statistics*. 1976;1(1):69-82. doi:10.3102/10769986001001069
5. Greenhouse SW, Geisser S. On methods in the analysis of profile data. *Psychometrika*. 1959;24(2):95-112. doi:10.1007/BF02289823
6. Blanca MJ, Arnau J, García-Castro FJ, Alarcón R, Bono R. Repeated measures ANOVA and adjusted tests when sphericity is violated: Which procedure is best? *Frontiers in Psychology*. 2023;14:1192453. doi:10.3389/fpsyg.2023.1192453
7. Kent DM, Paulus JK, Klaveren D van, et al. The Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement. *Annals of Internal Medicine*. 2020;172(1):35-45. doi:10.7326/M18-3667

- 1507 8. Kent DM, Klaveren D van, Paulus JK, et al. The PATH statement: Explanation and elaboration. *Annals of Internal Medicine*. 2020;172(1):W1-W25. doi:10.7326/M18-3668
- 1508 9. Senn S. The AB/BA crossover: Past, present and future? *Statistical Methods in Medical Research*. 1994;3(4):303-324. doi:10.1177/096228029400300402
- 1509 10. Senn S, Rolfe K, Julious SA. Investigating variability in patient response to treatment: A case study from a replicate cross-over study. *Statistical Methods in Medical Research*. 2011;20(6):657-666. doi:10.1177/0962280210379174
- 1510 11. Natesan Batley P, McClure EB, Brewer B, et al. Evidence and reporting standards in N-of-1 medical studies: A systematic review. *Translational Psychiatry*. 2023;13(1):263. doi:10.1038/s41398-023-02562-8
- 1511 12. Stunnenberg BC, Berends J, Griggs RC, et al. N-of-1 trials in neurology: A systematic review. *Neurology*. 2022;98(2):e174-e185. doi:10.1212/WNL.00000000000012998
- 1512 13. Shaffer JA, Kronish IM, Falzon L, Cheung YK, Davidson KW. N-of-1 randomized intervention trials in health psychology: A systematic review and methodology critique. *Annals of Behavioral Medicine*. 2018;52(9):731-742. doi:10.1093/abm/kax026
- 1513 14. Vohra S, Shamseer L, Sampson M, et al. CONSORT extension for reporting N-of-1 trials (CENT) 2015 statement. *Journal of Clinical Epidemiology*. 2015;76:9-17. doi:10.1016/j.jclinepi.2015.05.004
- 1514 15. Shamseer L, Sampson M, Bukutu C, Schmid CH, et al. CENT 2015: Explanation and elaboration. *Journal of Clinical Epidemiology*. 2015;76:18-46. doi:10.1016/j.jclinepi.2015.05.018
- 1515 16. Kwasnicka D, Inauen J, Nieuwenboom W, et al. Challenges and solutions for N-of-1 design studies in health psychology. *Health Psychology Review*. 2019;13(2):163-178. doi:10.1080/17437199.2018.1564627
- 1516 17. Mallinckrodt CH, Clark SWS, Carroll RJ, Molenberghs G. Assessing response profiles from incomplete longitudinal clinical trial data under regulatory considerations. *Journal of Biopharmaceutical Statistics*. 2003;13(2):179-190. doi:10.1081/BIP-120019265

- 1517 18. Mallinckrodt CH, Clark WS, David SR. Accounting for dropout bias using mixed-effects models. *Journal of Biopharmaceutical Statistics*. 2001;11(1-2):9-21. doi:10.1081/BIP-100104194
- 1518 19. Mallinckrodt CH, Raskin J, Wohlreich MM, Watkin JG, Detke MJ. The efficacy of duloxetine: A comprehensive summary of results from MMRM and LOCF_ANCOVA in eight clinical trials. *BMC Psychiatry*. 2004;4:26. doi:10.1186/1471-244X-4-26
- 1519 20. Prakash A, Risser RC, Mallinckrodt CH. The impact of analytic method on interpretation of outcomes in longitudinal clinical trials. *International Journal of Clinical Practice*. 2008;62(8):1147-1158. doi:10.1111/j.1742-1241.2008.01808.x
- 1520 21. Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13
- 1521 22. Wang M, Kong L, Li Z, Zhang L. Covariance estimators for generalized estimating equations (GEE) in longitudinal analysis with small samples. *Statistics in Medicine*. 2016;35(10):1706-1721. doi:10.1002/sim.6817
- 1522 23. Pan W. Sample size and power calculations with correlated binary data. *Controlled Clinical Trials*. 2001;22(3):211-227. doi:10.1016/s0197-2456(01)00131-3
- 1523 24. Tong G, Nevins P, Ryan M, others, Taljaard M. A review of current practice in the design and analysis of extremely small stepped-wedge cluster randomized trials. *Clinical Trials*. 2024;22(1):45-56. doi:10.1177/17407745241276137
- 1524 25. Hedges LV, Shadish WR, Natesan Batley P. Power analysis for single-case designs: Computations for $(AB)^k$ designs. *Behavior Research Methods*. 2023;55(7):3494-3503. doi:10.3758/s13428-022-01971-9
- 1525 26. Bouwmeester S, Jongerling J. Power of a randomization test in a single case multiple baseline AB design. *PLoS ONE*. 2020;15(2):e0228355. doi:10.1371/journal.pone.0228355
- 1526 27. Schork NJ, Beaulieu-Jones B, Liang WS, Smalley S, Goetz LH. Exploring human biology with N-of-1 clinical trials. *Cambridge Prisms: Precision Medicine*. 2023;1:e12. doi:10.1017/pcm.2022.15

- 1527 28. Zucker DR, Ruthazer R, Schmid CH. Individual (N-of-1) trials can be combined to give population comparative treatment effect estimates: Methodologic considerations. *Journal of Clinical Epidemiology*. 2010;63(12):1312-1323. doi:10.1016/j.jclinepi.2010.04.020
- 1528 29. Defelippe VM, Thiel GJMW van, Otte WM, Schutgens REG, Stunnenberg B, et al. Toward responsible clinical n-of-1 strategies for rare diseases. *Drug Discovery Today*. 2023;28(10):103688. doi:10.1016/j.drudis.2023.103688
- 1529 30. Berger VW, Rezvani A, Makarewicz VA. Direct effect on validity of response run-in selection in clinical trials. *Controlled Clinical Trials*. 2003;24(2):156-166. doi:10.1016/s0197-2456(02)00316-1
- 1530 31. Senn S. *Cross-over Trials in Clinical Research*. 2nd ed. Wiley; 2002.
- 1531 32. Maxwell SE, Delaney HD, Kelley K. *Designing Experiments and Analyzing Data: A Model Comparison Perspective*. 3rd ed. Routledge; 2018.
- 1532 33. Mallinckrodt CH, Lane PW, Schnell D, Peng Y, Mancuso JP. Recommendations for the primary analysis of continuous endpoints in longitudinal clinical trials. *Drug Information Journal*. 2008;42(4):303-319. doi:10.1177/009286150804200402
- 1533 34. Mauchly JW. Significance test for sphericity of a normal n-variate distribution. *Annals of Mathematical Statistics*. 1940;11(2):204-209. doi:10.1214/aoms/1177731915
- 1534 35. Mancl LA, DeRouen TA. A covariance estimator for GEE with improved small-sample properties. *Biometrics*. 2001;57(1):126-134.